

Net change in money



- 1 Describe what it means if the net change in money is:
 - a Positive
 - b Negative

- 2 For each of the following, describe the change in money:
 - a After buying and then selling a bicycle, the net change in Laura's money was $-\$44$.
 - b After buying and then selling a scarf, the net change in Sandy's money was $\$5$.
 - c After buying and then selling a chair, the net change in Derek's money was $-\$8$.
 - d After buying and then selling a scarf, the net change in Bianca's money was $\$0$.
 - e After buying and then selling a toaster, the net change in Edward's money was $\$65$.

- 3 For each of the following items, find the net change in money as a directed number:
 - a Valentina bought a pillow for $\$2$ and sold it for $\$16$.
 - b Bill bought a chair for $\$17$ and sold it for $\$6$.
 - c Hannah bought a tennis racquet for $\$82$ and sold it for $\$169$.
 - d Aoife bought a car for $\$12\,000$ and sold it five years later for $\$4500$.
 - e Yan bought an apartment for $\$380\,000$ and sold it ten years later for $\$520\,000$.

- 4 In its opening month, a local pizza store sold 190 large pizzas at $\$20$ each and 200 small pizzas at $\$6$ each. The store incurred electricity costs of $\$26$, water costs of $\$10$, rent of $\$408$, wages of $\$1345$ and ingredient costs of $\$210$.
 - a Calculate the total revenue of the store in the first month.
 - b Calculate the total expenses of the store in the first month.
 - c Calculate the net change in money by the store in the first month.



Cost and sale price

5 Find the sale price of the following items:

- a** Cost price: \$58, profit: \$10.
- b** Cost price: \$120, profit: \$32.50.
- c** Cost price: \$80, loss: \$27.
- d** Cost price: \$250, loss: \$135.
- e** Sandy bought a stove for \$238. She eventually sold the stove and made a profit of \$39.
- f** William bought a wallet for \$12. He eventually sold the wallet and made a profit of \$2.
- g** Pauline bought a plane ticket for \$469. She then had to sell the plane ticket and made a loss of \$9.
- h** Ryan bought a guitar for \$176. He eventually sold the guitar and made a loss of \$37.

6 Find the cost price of the following items:

- a** Sale price: \$72, profit: \$15.
- b** Sale price: \$10 000, profit: \$2200.
- c** Sale price: \$520, loss: \$210.
- d** Sale price: \$780, loss: \$360.
- e** Ben sold a toothbrush for \$12 and made a profit of \$3.
- f** Pauline sold a washing machine for \$486 and made a loss of \$7.
- g** Sarah sold a laptop for \$233 and made a profit of \$55.
- h** David sold a pair of earrings for \$161 and made a loss of \$55.

7 Find the profit on the following items:

- a** Cost price: \$65, sale price: \$82.
- b** Cost price: \$120, sale price: \$186.
- c** Cost price: \$1400, sale price: \$2050.
- d** Cost price: \$86.50, sale price: \$93.50.
- e** Glen bought a trading card for \$11 and sold it for \$24.
- f** Roxanne bought a dress for \$712 and sold it for \$829.

8 Find the loss on the following items:



- a** Cost price: \$70, sale price: \$45.
- b** Cost price: \$225, sale price: \$138.
- c** Cost price: \$5870, sale price: \$4250.
- d** Cost price: \$850, sale price: \$375.50.
- e** Judy bought a basketball for \$99 and sold it for \$92.
- f** Charlie bought a sculpture for \$660 and sold it for \$585.

9 Find the profit (or loss) for each of the following:

- a** The sale price is \$540 and the cost price is \$284.
- b** The sale price is \$2065 and the cost price is \$3120.
- c** The sale price is \$340.50 and the cost price is \$290.80.
- d** The sale price is \$1800 and the cost price is \$3500.
- e** The cost price is \$276.90 and the sale price is \$320.55.
- f** The cost price is \$2480 and the sale price is \$1770.

Percentage profit and loss

- 10** Charlie bought a cake for \$220 and sold it for \$209. Describe Charlie's change in money after buying and selling the cake.
- 11** Sean bought a spade for \$36 and sold it for \$45. Describe Sean's change in money after buying and selling the spade.
- 12** Oliver bought a baseball bat for \$130 and sold it for \$143.
 - a** Calculate Oliver's profit.
 - b** Find the percentage profit of the cost price of the baseball bat.
 - c** Describe Oliver's change in money after buying and selling the baseball bat.
- 13** Bianca bought a microwave for \$50 and sold it for \$35.
 - a** Calculate Bianca's loss.
 - b** Find the percentage loss of the cost price of the microwave.
 - c** Describe Bianca's change in money after buying and selling the microwave.

14 John buys a used laptop for \$391 and resells it for \$554.



- a** Calculate the profit John make on the laptop.
- b** Find this profit as a percentage of the cost price.

15 Paul invests \$9000 in the stock market only to see his shares plummet in value to \$4509.

- a** Calculate the loss Paul make on the shares.
- b** Find this loss as a percentage of the cost price.

16 Sally purchased a dress for \$20 and auctioned it off on eBay for \$41. Find:

- a** The profit (or loss).
- b** The profit (or loss) as a percentage of the cost, correct to two decimal places.
- c** The profit (or loss) as a percentage of the sale price to two decimal places.

17 Buzz bought a mobile phone for \$170 and sold it 7 months later, making a loss of 30% on the purchase price. Find the selling price mobile phone.

18 Every item in store is sold at 8% above cost. If a product was bought from the manufacturer for \$360, find:

- a** The sale price
- b** The profit (or loss)

19 Sarah bought a bicycle for \$400 and sold it for a 18% profit.

- a** Find the profit made on the bicycle.
- b** Hence, find the sale price of the bicycle.

20 Hermione bought a phone for \$200 and sold it for a 16% loss.

- a** Find the loss made on the phone.
- b** Hence, find the sale price of the phone.

21 Noah sold a jumping castle for \$700 which was a 12% profit.

- a** Find the sale price of the jumping castle as a percentage of the cost price.
- b** Hence, find the cost price of the jumping castle.

22 James sold a jacket for \$600 which was a 4% loss.

- a** Find the sale price of the jacket as a percentage of the cost price.
- b** Hence, find the cost price of the jacket.



23 Yvonne bought a car for \$17 600 and sold it  years later, making a loss of 19%. Find the sale price of the car.